

# Documentation of the pipeline

September 12, 2026

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## 1 Data

- data/calderon\_data/new\_time/NN/  
Single-cell ATAC-seq counts in cells from embryonic development of *Drosophila melanogaster*. For each time window there are files with peak coordinates, cell barcodes and the count matrix
- data/calderon\_data/atac\_meta.rds  
Metadata for the cells from scATAC-seq experiment
- data/calderon\_data/atac\_meta.readme.txt  
Metadata readme (description of metadata columns)
- data/calderon\_data/science.abn5800\_table\_s2.xlsx  
Lineage info for clusters of cells
- data/long\_and\_short\_range\_loops\_D\_mel.tsv  
Chromatin loops coordinates for the *Drosophila melanogaster* organism

## 2 Preparation of metadata

tw  $\in \{00-02, 02-04, \dots, 18-20\}$

**source code:** src/R/prepare\_cluster\_cardinality\_info.R

**results:** results/cluster\_cardinality.tsv.gz

Columns: seurat\_clusters\_predtime, {tw}

**source code:** src/R/prepare\_cluster\_population\_info.R

**results:** results/cluster\_pop\_mapping.tsv.gz

Columns: time\_window, cluster\_id, refined\_annotation, population

**source code:**

src/R/prepare\_loop\_activity\_info.R

**results:** results/loop\_activity\_info.tsv

Columns: loop\_id,

Dmel\_6-8h\_Neuroblasts, Dmel\_6-8h\_Neurons, Dmel\_6-8h\_Glia,

Dmel\_10-12h\_Neuroblasts, Dmel\_10-12h\_Neurons, Dmel\_10-12h\_Glia,

Dmel\_14-16h\_Neuroblasts, Dmel\_14-16h\_Neurons, Dmel\_14-16h\_Glia,

Dmel\_16-18h\_whole\_embryo, Dmel\_larval\_brain, Dmel\_adult\_brain,

small\_profile (concatenated first 9 binary activity columns),

large\_profile (concatenated 12 binary activity columns)

### 3 Preparation of total reads and scATAC-seq counts in chromatin loop anchors

$tw \in \{00-02, 02-04, \dots, 18-20\}$

**source code:**

src/R/prepare\_total\_reads\_info.R

src/snakemake/prepare\_total\_reads\_info.Snakefile

**results:** results/counts/total\_reads/total\_reads\_{tw}.tsv.gz

Columns: barcode, cluster, annotation, population, total\_reads, number\_of\_peaks

**source code:**

src/R/prepare\_counts\_in\_anchors.R

src/snakemake/prepare\_counts\_in\_anchors.Snakefile

**results:** results/counts/counts\_in\_anchors/reads\_{tw}\_{loop\_id}.tsv.gz

Columns: barcode, population, total\_reads, x\_A1, x\_A2

### 4 Estimation of parameters of distributions of scATAC-seq counts

**source code:**

src/R/estimate\_beta\_distribution\_parameters\_of\_counts\_in\_anchors.R

src/snakemake/estimate\_beta\_distribution\_parameters\_of\_counts\_in\_anchors.Snakefile

**results:** results/parameters/beta\_distribution\_of\_counts\_in\_anchors/

- beta\_parameters\_{tw}\_{loop\_id}\_all\_and\_by\_population.tsv.gz

- log\_file\_tw\_{tw}.txt

- logs/{tw}\_{loop\_id}.log

**source code:**

src/R/generate\_beta\_density\_report.R

Command-line arguments: {loop\_id}, {xmin}, {xmax} (interval for density plotting)

**results:**

results/parameters/beta\_distribution\_of\_counts\_in\_anchors\_plots/report\_beta\_densities\_loop\_{loop\_id}.pdf

### 5 Estimation of parameters of distributions of total reads

**source code:**

src/R/estimate\_negbinom\_distribution\_parameters\_of\_total\_reads.R

src/snakemake/estimate\_negbinom\_distribution\_parameters\_of\_total\_reads.Snakefile

**results:** results/parameters/negbinom\_distribution\_of\_total\_reads/

- negbinom\_parameters\_{tw}\_all\_and\_by\_population.tsv.gz

- log\_file\_tw\_{tw}.txt

- logs/{tw}.log

**source code:** src/R/generate\_negbinom\_density\_report.R

**results:** results/parameters/negbinom\_distribution\_of\_total\_reads/report\_negbinom\_densities\_with\_histograms.pdf

## 6 Synthetic data generation - large grid of parameters, only 1 dataset for each combination (refined in further steps)

**source code:**

src/R/generate\_synthetic\_data.R

src/snakemake/generate\_synthetic\_data.Snakefile

---

RHOS = [-0.6, 0.0, 0.6]

N\_CELLS = [ 500, 1000, 1500, 2000, 2500, 3000, 3500, 4000, 5000, 10000, 15000, 20000, 25000, 30000, 35000, 40000, 45000, 50000, 55000, 60000, 80000, 100000, 120000, 140000, 160000 ]

ALPHA = [0.15] BETA = [600]

MUS = [1000, 2000, 3000, 4000, 5000, 6000]

"fixed" = all total reads equal to mu, "inf" = Poisson

SIZE\_NEGBINOMS = ["inf", "fixed", "0.5", "1.0", "1.5", "2.0", "2.5", "3.0", "3.5", "10"]

---

**results:** results/synthetic\_data/

- rho\_{rho}/synthetic\_counts\_n{n}\_mu{mu}\_sizeNB{size\_nb}\_alpha{alpha}\_beta{beta}.tsv.gz

Columns:

barcode, total\_reads, x\_A1, x\_A2, sigma, generates\_norm\_cor, prob\_pearson\_cor, prob\_spearman\_cor

- logs/rho\_{rho}/sim\_n{n}\_mu{mu}\_sizeNB{size\_nb}\_alpha{alpha}\_beta{beta}.log

## 7 Extended synthetic data generation - larger grid of parameters than in 'Synthetic data generation'. We generate 20 datasets for each combination. For some combinations (subset of a grid) we generate 1000 datasets. Final version of parameter grid with 1000 datasets for each combination will be introduced in the next step

**source code:**

src/R/generate\_synthetic\_data\_extended.R

src/snakemake/generate\_synthetic\_data\_extended.Snakefile

---

First version:

RHOS = [-1.0, -0.9, -0.8, -0.7, -0.6, -0.5, -0.4, -0.3, -0.2, -0.1, 0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]

SEEDS = list(range(20))

Large grid of parameters:

n\_cells\_list <- c( 500, 1000, 1500, 2000, 2500, 3000, 3500, 4000, 5000, 10000, 20000, 40000, 60000, 80000, 100000, 120000, 140000, 160000 )

```
mus_list <- c(1000, 2000, 3000, 4000, 5000, 6000)
size_nb_strings <- c("inf", "fixed", "0.1", "0.2", "0.3", "0.4", "0.5", "0.6", "0.7", "0.8", "0.9", "1.0", "1.5",
"2.0", "2.5", "3.0", "3.5", "10.0")
alpha_val <- 0.15
beta_val <- 600
```

For each rho from [-1.0, -0.9, -0.8, -0.7, -0.6, -0.5, -0.4, -0.3, -0.2, -0.1, 0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0] and each combination of parameters from this large grid there are 20 datasets (from different seeds).

---

Second version:

```
RHOS = [-0.6, 0.0, 0.6]
SEEDS = list(range(1000))
Small grid of parameters:
n_cells_list <- c(1000, 2000, 5000, 10000, 20000)
mus_list <- c(5000)
size_nb_strings <- c("inf", "fixed", "0.5", "1.0", "2.0")
alpha_val <- 0.15
beta_val <- 600
```

For each rho from [-0.6, 0.0, 0.6] and each combination of parameters from this small grid there are 1000 datasets (from different seeds).

---

**results:** results/synthetic\_data\_extended/

- rho\_{rho}/synthetic\_counts\_n{n}\_mu{mu}\_sizeNB{size\_nb}\_alpha{alpha}\_beta{beta}\_seed{seed}.tsv.gz

Columns:

total\_reads, x\_A1, x\_A2, sigma, generates\_norm\_cor, prob\_pearson\_cor, prob\_spearman\_cor

- rho\_{rho}/sim\_seed{seed}.done

- logs/rho\_{rho}/sim\_seed{seed}.log

## 8 Extended synthetic data generation plus - final grid of parameters (not too large, but not too small). We generate 1000 datasets (with distinct seeds) for each combination of parameters

**source code:**

src/R/generate\_synthetic\_data\_extended\_plus.R

src/snakemake/generate\_synthetic\_data\_extended\_plus.Snakefile

**results:**

results/synthetic\_data\_extended\_plus/rho\_{rho}/synthetic\_counts\_seed{seed}.tsv.gz

Columns:

n\_cells, mu, size\_nb, total\_reads, x\_A1, x\_A2, sigma, generated\_norm\_cor, prob\_pearson\_cor, prob\_spearman\_cor

results/synthetic\_data\_extended\_plus/logs/rho\_{rho}/sim\_seed{seed}.log

---

Parameter grid:

RHOS = [-0.6, -0.5, -0.4, -0.3, -0.2, -0.1, 0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6]

SEEDS = list(range(1000))

n\_cells\_list <- c(1000, 2000, 5000, 10000, 20000)

mus\_list <- c(1000, 2000, 3000, 4000, 5000)

size\_nb\_strings <- c("0.1", "0.2", "0.5", "1.0", "2.0", "inf", "fixed")

alpha\_val <- 0.15

beta\_val <- 600

## 9 Distribution fitting for synthetic and real data

### source code:

src/R/MGLMfit\_GDM.R

src/snakemake/MGLMfit\_GDM\_synthetic.Snakefile

src/snakemake/MGLMfit\_GDM\_real\_all.Snakefile

src/snakemake/MGLMfit\_GDM\_real\_pops.Snakefile

src/bash/MGLMfit\_GDM\_real\_pops\_by\_tw\_and\_pop.sh

### results:

For  $\text{init} \in \{\text{defalut}, 1e-4, 1e-5, 1e-6, 1e-7, 1e-8\}$

results/MGLMfit\_GDM/synthetic\_data/rho\_{rho}/init\_{init}/

- fit\_synthetic\_n{n}\_mu{mu}\_sizeNB{size\_nb}\_alpha{alpha}\_beta{beta}.rds

- fit\_GDM.log

For  $\text{init} \in \{\text{defalut}, 1e-6\}$

results/MGLMfit\_GDM/real\_data/all/init\_{init}/

- fit\_real\_{tw}\_{loop\_id}.rds

- fit\_GDM\_{tw}.log

For  $\text{init} \in \{\text{defalut}, 1e-6\}$

results/MGLMfit\_GDM/real\_data/pops/init\_{init}/

- fit\_real\_{tw}\_{loop\_id}\_pop\_{pop\_id}.rds

- fit\_GDM\_{tw}\_pop\_{pop\_id}.log

### source code:

src/R/MGLMfit\_GDM\_extended.R

src/snakemake/MGLMfit\_GDM\_synthetic\_extended.Snakefile

### results:

For  $\text{init} \in \{1e-6\}$

results/MGLMfit\_GDM/synthetic\_data\_extended/rho\_{rho}/init\_{init}/

- fit\_synthetic\_n{n}\_mu{mu}\_sizeNB{size\_nb}\_alpha{alpha}\_beta{beta}\_seed{seed}.rds

- fit\_seed{seed}.done

- fit\_GDM\_shared.log

results/MGLMfit\_GDM/synthetic\_data\_extended/logs/rho\_{rho}/

- fit\_rho\_{rho}\_init\_{init}\_seed{seed}.log

**source code:**

src/R/MGLMfit\_GDM\_extended\_plus.R

src/snakemake/MGLMfit\_GDM\_synthetic\_extended\_plus.Snakefile

**results:**

For  $\text{init} \in \{1e-6\}$

results/MGLMfit\_GDM/synthetic\_data\_extended\_plus/rho\_{rho}/init\_{init}/

- fit\_synthetic\_n{n}\_mu{mu}\_sizeNB{size\_nb}\_alpha{alpha}\_beta{beta}\_seed{seed}.rds

- fit\_seed{seed}.done

- fit\_GDM\_shared.log

results/MGLMfit\_GDM/synthetic\_data\_extended\_plus/logs/rho\_{rho}/

- fit\_rho\_{rho}\_init\_{init}\_seed{seed}.log

Although synthetic data extended plus was generated for

$\text{RHOS} = [-0.6, -0.5, -0.4, -0.3, -0.2, -0.1, 0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6]$ ,

MGLMfit procedure for synthetic data extended plus is performed for

$\text{RHOS} = [-0.6, -0.4, -0.2, 0.0, 0.2, 0.4, 0.6]$ ,

as we decided that it is sufficient for our analysis.

## 10 Checking results of distribution fitting and computing correlation between chromatin accessibility at chromatin loop anchors

**source code:**

src/R/MGLMfit\_GDM\_cor/

- correlation\_functions.R

- cor\_synthetic.R

- cor\_real\_all.R

- cor\_real\_pops.R

- cor\_synthetic\_extended.R

- cor\_synthetic\_extended\_plus.R

src/snakemake/MGLMfit\_GDM\_cor/

- cor\_synthetic.Snakefile

- cor\_real\_all.Snakefile

- cor\_real\_pops.Snakefile

- cor\_synthetic\_extended.Snakefile

- cor\_synthetic\_extended\_plus.Snakefile

**results:**

For  $\text{init} \in \{\text{defalut}, 1e-4, 1e-5, 1e-6, 1e-7, 1e-8\}$

results/MGLMfit\_GDM\_cor/synthetic\_data/rho\_{rho}/init\_{init}/cor.tsv.gz

results/MGLMfit\_GDM\_cor/synthetic\_data/logs/cor\_rho\_{rho}\_init\_{init}.log

For  $\text{init} \in \{\text{defalut}, 1e-6\}$

```

results/MGLMfit_GDM_cor/real_data/all/init_{init}/cor_{tw}.tsv.gz
results/MGLMfit_GDM_cor/real_data/all/logs/cor_{tw}_init_{init}.log

For init ∈ {defalut, 1e-6}
results/MGLMfit_GDM_cor/real_data/pops/init_{init}/cor_{tw}.tsv.gz
results/MGLMfit_GDM_cor/real_data/pops/logs/cor_{tw}_init_{init}.log

For init ∈ {1e-6}
results/MGLMfit_GDM_cor/synthetic_data_extended/rho_{rho}/init_{init}/cor_seed{seed}.tsv.gz
results/MGLMfit_GDM_cor/synthetic_data_extended/logs/rho_{rho}/cor_rho_{rho}_init_{init}_seed{seed}.log

For init ∈ {1e-6}
results/MGLMfit_GDM_cor/synthetic_data_extended_plus/rho_{rho}/init_{init}/cor_seed{seed}.tsv.gz
results/MGLMfit_GDM_cor/synthetic_data_extended_plus/logs/rho_{rho}/cor_rho_{rho}_init_{init}_seed{seed}.log

```

## 11 Visualization of computed correlations for synthetic and real data

### source code:

```

src/R/MGLMfit_GDM_cor/plot_real_correlations.R
src/R/MGLMfit_GDM_cor/plot_activity_correlations.R
src/R/MGLMfit_GDM_cor/plot_synthetic_correlations.R
src/R/MGLMfit_GDM_cor/plot_synthetic_extended_gdm.R
src/snakemake/MGLMfit_GDM_cor/plot_real_correlations.Snakefile
src/snakemake/MGLMfit_GDM_cor/plot_activity_correlations.Snakefile
src/snakemake/MGLMfit_GDM_cor/plot_synthetic_correlations.Snakefile
src/snakemake/MGLMfit_GDM_cor/plot_synthetic_extended_gdm.Snakefile

```

### results:

```

results/MGLMfit_GDM_cor/real_data/all/init_{init}/plots_report.pdf
results/MGLMfit_GDM_cor/real_data/all/init_{init}/plots_activity_report.pdf
results/MGLMfit_GDM_cor/real_data/pops/init_{init}/plots_report.pdf
results/MGLMfit_GDM_cor/real_data/pops/init_{init}/plots_activity_report.pdf
results/MGLMfit_GDM_cor/synthetic_data/rho_{rho}/init_{init}/plots_report.pdf
results/plots_extended/gdm_correlations/rho_{rho}/init_{init}/
- variable_N.pdf
- variable_MU.pdf
- variable_SIZE.pdf

results/MGLMfit_GDM_cor/real_data/all/logs/
- plotting_init_{init}.log
- plotting_activity_init_{init}.log
results/MGLMfit_GDM_cor/real_data/pops/logs/
- plotting_init_{init}.log
- plotting_activity_init_{init}.log
results/MGLMfit_GDM_cor/synthetic_data/logs/plotting_rho_{rho}_init_{init}.log
results/plots_extended/gdm_correlations/logs/plotting_rho_{rho}_init_{init}.log

```

## 12 Generalized linear model fitting for real data

### source code:

src/R/MGLMreg/add\_NNv1\_age\_to\_reads.R

src/R/MGLMreg/

- GDM\_unified\_reg.R

- check\_unified\_reg.R

- estimates\_unified\_reg.R

- wald\_unified\_reg.R

src/snakemake/MGLMreg/

- GDM\_unified\_reg.Snakefile

## 13 Interpretation of generalized linear model parameters for real data

### source code:

src/R/MGLMreg/eval\_mglmreg\_window.R

src/R/MGLMreg/eval\_mglmreg\_window\_tissue.R

src/snakemake/MGLMreg/eval\_mglmreg\_window.Snakefile

src/snakemake/MGLMreg/eval\_mglmreg\_window\_tissue.Snakefile

## 14 Visualization of distribution fitting results for real data

### source code:

src/R/triangle\_plots/for\_thesis/plot\_triangle\_dens\_all.R

src/R/triangle\_plots/for\_thesis/plot\_triangle\_dens\_pops.R

## 15 Motivating example

### source code:

src/R/MGLMfit\_GDM\_cor/naive\_cor\_real.R

src/R/MGLMfit\_GDM\_cor/naive\_cor\_synthetic.R

src/R/MGLMfit\_GDM\_cor/naive\_cor\_synthetic\_extended.R

src/R/snakemake/MGLMfit\_GDM\_cor/naive\_correlations.Snakefile

src/R/snakemake/MGLMfit\_GDM\_cor/naive\_correlations\_extended.Snakefile

### results:

results/naive\_correlations/real\_data/all/cor\_{tw}.tsv.gz

results/naive\_correlations/real\_data/pops/cor\_{tw}.tsv.gz

results/naive\_correlations/synthetic\_data/rho\_{rho}/cor.tsv.gz

results/naive\_correlation/synthetic\_data\_extended/rho\_{rho}/naive\_cor\_seed{seed}.tsv.gz

results/naive\_correlations/logs/

- naive\_real\_all\_{tw}.tsv.gz
- naive\_real\_pops\_{tw}.tsv.gz
- naive\_synthetic\_rho\_{rho}.tsv.gz

results/naive\_correlation/synthetic\_data\_extended/logs/cor\_rho\_{rho}\_seed{seed}.log

**source code:**

src/R/MGLMfit\_GDM\_cor/plot\_naive\_real.R  
 src/R/MGLMfit\_GDM\_cor/plot\_naive\_synthetic.R  
 src/R/MGLMfit\_GDM\_cor/plot\_synthetic\_extended\_naive.R  
 src/snakemake/MGLMfit\_GDM\_cor/plot\_naive\_correlations.Snakefile  
 src/snakemake/MGLMfit\_GDM\_cor/plot\_synthetic\_extended\_naive.Snakefile

**results:**

results/naive\_correlations/real\_data/all/plots\_report\_naive.pdf  
 results/naive\_correlations/real\_data/pops/plots\_report\_naive.pdf  
 results/naive\_correlations/synthetic\_data/rho\_{rho}/plots\_report\_naive.pdf  
 results/plots\_extended/naive\_correlations/rho\_{rho}/  
 - variable\_N.pdf  
 - variable\_MU.pdf  
 - variable\_SIZE.pdf

results/naive\_correlations/real\_data/all/logs/plotting\_naive.log  
 results/naive\_correlations/real\_data/pops/logs/plotting\_naive.log  
 results/naive\_correlations/synthetic\_data/logs/plotting\_rho\_{rho}.log  
 results/plots\_extended/naive\_correlations/logs/plotting\_rho\_{rho}.log

## 16 Plots for a poster

**source code:**

src/R/plots\_for\_a\_poster/  
 - nb\_parameter\_exploration.R  
 - plot\_synthetic\_naive.R  
 - plot\_synthetic\_gdm.R  
 - plot\_real\_activity.R  
 - extract\_one\_loop\_correlation.R

**results:**

nb\_parameter\_exploration\_R.pdf  
 naive\_matrix\_pearson.pdf  
 naive\_matrix\_spearman.pdf  
 gdm\_matrix\_by\_stability.pdf  
 real\_data\_activity\_matrix.pdf  
 results/MGLMfit\_GDM\_cor/real\_data/pops/  
 - plot\_{loop\_id}\_spearman\_lines\_00\_10.pdf  
 - plot\_{loop\_id}\_spearman\_lines\_10\_20.pdf

## 17 Lineage analysis

### source code:

```
src/R/lineage_analysis/  
- prepare_lineage_info.R  
- prepare_tree.R  
- prepare_indented_tree.R  
- plot_all_loops.R  
- plot_each_loop_separately.R  
- plot_all_loops_as_points.R  
- plot_all_loops_by_activity.R  
- plot_all_loops_by_activity_3_windows.R
```

### results:

```
results/lineage_analysis/  
- lineage_populations.tsv  
- time_window_clusters.txt  
- cluster_tree_indented.txt  
- lineage_only_tree_indented.txt  
- lineage_parents_mapping.tsv  
- root_to_leaf_paths.txt  
- correlation_by_lineage.pdf  
- correlation_by_loop.pdf  
- correlation_by_lineage_point_for_each_loop.pdf  
- correlation_by_lineage_with_activity.pdf  
- correlation_by_lineage_with_union_activity_3_windows.pdf
```

## 18 Plots for a thesis

### source code:

```
src/R/plots_for_a_thesis/  
- reads_statistics.R  
- synthetic_correlations/*  
- real_correlations/*  
- lineage_analysis/*  
- loops_statistics/*  
- MGLMreg_analysis/*  
- parameters/*
```